THE ORBITAL ELECTRON IN THE TORSION MEDIUM

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ABSTRACT

The electron (E+ spinor irrep of the 2I double group, vertex mode of the torsion medium) experiences four geometrically distinct forces when in orbit around a proton. This paper identifies each force, its physical mechanism, its torsionverse derivation status, and its contribution to orbital structure. The key result: the electron's roll direction (the chirality of its (1,2) Hopf winding) is not merely a labelling convention but determines the sign of its electromagnetic coupling to the proton. A positron (E- spinor irrep of 2I, identical vertex geometry to E+ but opposite roll) is repelled from the proton for the same geometric reason that two protons repel. The four orbital forces together produce atomic shell quantization from I_h icosahedral geometry with zero free parameters. Additionally: at each vertex contact the (1,2) Hopf winding splits force exactly 2:1 forward-to-sideways (Section 2.3); positronium annihilation rules out classical-amplitude wave models and requires discrete winding energy, consistent with both QFT and the torsionverse corpuscle picture (Section 5.3.1).

1. INTRODUCTION

This paper covers orbital mechanics only. Mass derivation is not in scope here -- see [doc_leptons] (LM1-LM2), [doc_alpha] (Born balance), [doc_jobson_cell] (J26).

The electron (E+ spinor irrep, vertex mode of the torsion medium) is one of the three fundamental lepton modes of the torsionverse framework. This paper derives its four-force orbital structure around the proton, the physical origin of charge sign from winding chirality, and the Z > 137 Coulomb stability limit from corpuscle circuit geometry.

Section 2 identifies the electron as the E+ vertex mode, establishes its cell-interaction regime, and derives the winding force decomposition (2:1 forward-to-sideways ratio from (1,2) geometry). Section 3 derives the four geometrically distinct forces on the orbital electron (pressure, light linger, roll, wall). Section 4 shows that roll direction determines charge sign, derived from cell-meshing mechanics. Section 5 compares the electron and positron. Section 6 derives Z_crit = 1/alpha from corpuscle circuit geometry. Section 7 summarises the force table and the roll-charge relationship. Section 8 lists open items.

Companion scripts: analysis/demos/electron_doc.py (19/19 PASS, ED1-ED19) [STANDALONE -- no external deps]; see the companion scripts section below.

2. THE ELECTRON AS E+ VERTEX MODE

2.1 Irrep assignment

The icosahedral double group 2I has four spinor irreps (half-integer spin):

```
E+ (dim=2), E- (dim=2), G32 (dim=4), I52 (dim=6)
```

The electron is assigned to E+ on two grounds:

```
(1) Spin-1/2: dim=2 spinor, consistent with measured g-factor [doc_alpha]
(2) Vertex mode: E+ winding scatters at the 12 icosahedral vertices.
    chi(E+, C5) = +phi  (phi = (1+sqrt(5))/2 = 1.618; constructive at 5-fold vertex nexuses).
    chi(E+, C3) = -1   (color singlet: vertex where 5 colored faces meet).
```

E- is the positron. Both E+ and E- have the same real C5 character (chi(E±,C5) = +phi), because phi is real and is its own complex conjugate. The distinction between E+ and E- is roll direction relative to the proton's Zone 2 chirality reference: E+ meshes (electron), E- grinds (positron). Same vertex geometry, reversed chirality. See Section 5.

2.2 Cell interaction regime

```
N_J_e = hbar*c / (m_e * c^2 * L_J) = 38,870  [deep bulk]
```

The electron is deep in the bulk regime (N_J >> 1). It interacts with the Jobson cell lattice through vertex contacts (the icosahedral vertex nexuses), not through excluded-volume displacement. There is no Zone 1 or Zone 2 for the electron -- its displacement is topological (winding number 2 of the (1,2) Hopf torus), not geometric (no nuclear hard core on the attraction side).

This is why the electron-proton system has no hard-core repulsion: cells MESH at the electron-proton interface (complementary chirality), whereas two protons GRIND (same chirality, Zone 2 contact at r = 2*lambda_p = 0.421 fm;

```
lambda_p = r_p/4 = 0.21 fm is the Zone 1/2 boundary radius).
```

[doc_nucleus Section 6; doc_orbit_pressure Section 1.2]

2.3 Winding force decomposition

The (1,2) Hopf torus path makes angle theta = arctan(p/q) = arctan(1/2) with the forward direction at each of the 12 vertex contacts per winding:

```
cos(theta) = q/||v|| = 2/sqrt(5)  (forward fraction -- maintains c-propagation)
sin(theta) = p/||v|| = 1/sqrt(5)  (sideways fraction -- displaces cells)
cot(theta) = q/p = 2/1 = 2        (EXACT integer ratio from winding numbers)
```

At every vertex contact the interaction force splits exactly 2:1, forward to sideways. The sideways component is what displaces cells and builds the 1/r^2

Coulomb pressure field described in Section 3.1. The corpuscle travels at c along the winding path itself; the forward (cos*c) and sideways (sin*c) components are projections of that path speed onto the two directions, with geometric sum = c. This 2:1 ratio is universal for all (1,2) corpuscles regardless of lepton generation -- the absolute energy scale differs (m_e, m_mu, m_tau) but the 2:1 force split is fixed by the winding geometry. [corpuscle_energy_decomp.py CE1-CE2 PASS]

3. THE FOUR ORBITAL FORCES

The electron in orbit around a proton is subject to four geometrically distinct mechanisms. These are not separate fundamental forces -- all four emerge from the same torsion medium physics. The labels name where each effect is dominant.

3.1 PRESSURE (primary driving force)

```
MECHANISM — PROTON GRADIENT:
  The proton's (1,2) Hopf winding in Zone 2 freezes Jobson cells into a
  specific chirality, expanding and thinning the medium in Zone 3. This
  creates a pressure gradient: Zone 3 cells are sparser (lower pressure)
  than the background. Any winding mode accumulates longer dwell times
  heading into the lower-pressure region and shorter dwell times heading
  out. The net drift is inward. This is gravity and electromagnetism --
  not two forces but one mechanism: dwell-time asymmetry in a pressure
  gradient. [doc_orbit_pressure Sections 1.1-1.2]
  The coupling strength differs because the source geometry differs:
  EM couples via surface winding (exponent 1), gravity couples via
  volumetric exclusion (18-dimensional Maxwell geometry; [doc_jobson_cell]). The ratio
  alpha_em/alpha_grav = 1.24e36 is derived, not assumed.
  [doc_orbit_pressure Section 1.2; orbit_doc.py OD6, OD11-OD14 PASS]

MECHANISM — ELECTRON SELF-WAKE:
  The electron bumps into every Jobson cell it encounters — it does not
  clear a path ahead. At each vertex contact (chi(E+,C5) = phi, Born
  coupling alpha) the electron knocks the cell out of the way as it
  passes through. Displaced cells haven't snapped back yet, leaving:
    - LOW pressure in the WAKE (tail) behind the electron: cells just
      passed through are displaced and sparse.
    - HIGHER pressure ahead and to the sides: unperturbed cells the
      electron is about to hit, and cells pushed laterally.
  The fractional displacement per contact is alpha -- the fine structure
  constant quantifies how hard each collision displaces a cell
  (k_n/k_eff from the Born balance, [doc_alpha] J17/J24).

  The orbital path curves inward because the proton's Zone 3 gradient
  makes the inward arc additionally sparse: cells are already thinned
  inward, so the electron accumulates longer dwell times going inward
  and is steered that way. The orbit is the electron perpetually following
  the inward-bending path of least resistance through unperturbed medium.

RESULT (Bohr radius):
  a_0 = hbar*c / (m_e * c^2 * alpha_em) = 5.292e-11 m  (exact, zero free parameters)
  DERIVED from an explicit balance of this section's own outward wave-packet
  confinement force against the proton's inward Coulomb pull -- not merely
  the formula plugged in: analysis/nuclear/bohr_radius_pressure_balance.py,
  8/8 PASS. (Corrects a prior citation here of "doc_orbit_pressure Section
  1.3; PO2 PASS" -- PO2 is Earth/Sun orbital mechanics, unrelated to this
  electron/proton result; doc_orbit_pressure Section 1.3 itself also only
  asserted the formula, not this balance-point derivation.)

Dominant over all other forces at atomic scales. Sets the orbital radius.
```

3.2 LIGHT LINGER (same pressure, wave-level readout)

```
MECHANISM:
  The low-pressure wake trailing behind the electron is sparser than
  the surrounding medium. Waves propagating back into the wake travel
  through this sparse region and linger there -- they take longer per cell
  hop than waves propagating forward (into the denser unperturbed medium
  ahead). This asymmetry means wave energy tends to remain in the wake
  behind the electron rather than radiate symmetrically in all directions.

  Simultaneously, the proton's Zone 3 gradient makes the inward-bending
  wake sparser still: the proton's own thinning reinforces the wake along
  the inward arc. Light lingers even longer along the inward path than the
  outward path, producing the net inward phase drift that we call the
  Coulomb potential in QM.

NOTE: light linger is not separate from the pressure mechanism. It is
what waves experience as they propagate through the pressure geometry --
the static description (pressure gradient) and the dynamic description
(wave linger asymmetry) are two views of the same physical situation.
```

3.3 ROLL (chirality coupling, orbital angular momentum)

```
MECHANISM:
  The Zone 2 Jobson cells are jammed (Maxwell-critical) and frozen in the
  proton's (1,2) Hopf chirality. They rotate with the proton. At each
  shared face boundary, the rotating Zone 2 cell drives the adjacent Zone 3
  cell through direct contact at the triangular face edges -- the gluon
  antinodes at the shared edge midpoints act as the coupling interface,
  like cogs on the triangular face boundaries. This propagates outward
  cell-by-cell: each Zone 3 cell, once driven, drives its neighbor at
  the next shared face edge. The mechanism is geometric and mechanical.
  [cell_rotation_propagation.py RP1-RP4:
  phi*L_J = 2*r_mid exactly (cells touch at antinodes), C3=+1 gluon
  coupling constructive, 72 deg forced by C5 geometry, propagates to
  all 12 vertices equally with zero attenuation]

  The 12-vertex icosahedral geometry restricts which angular momentum states
  are stable: the electron's standing wave in the Coulomb well decomposes
  into I_h irreps. The orbital quantum number l maps to irrep dimension:
    l=0: A_g (1)   s orbital
    l=1: T_1g (3)  p orbitals
    l=2: H_g (5)   d orbitals
    l=3: T_2g+G_g  f orbitals
  Degeneracy 2l+1 is exact for l=0..5 from I_h character theory [atomic_shells.py AS1].
  Shell maxima 2n² follow directly. Noble gas closings Z=2,10,18,36,54,86. [AS7]

  At larger orbital radii (higher n), the Zone 3 co-rotation field weakens
  as 1/r^3 (Lense-Thirring frame drag from the Hopf winding):
    E_Z3(r) = alpha * hbar*c * r_p^2 / r^3
  At r = r_p (Zone 3 onset): E_Z3 = alpha*hbar*c/r_p = 1.71 MeV.
  At r = a_0 (Bohr radius):  E_Z3 = 6.87 neV  (well below spin-orbit scale).
  [doc_entanglement; entanglement_doc.py EP1-EP2; electron_doc.py ED17-ED18 PASS]
  The 1/r^3 falloff is steeper than the pressure gradient (1/r^2), so roll
  becomes negligible at large n faster than the Coulomb well itself does,
  producing weaker spin-orbit effects for outer electrons.
```

3.4 WALL (inner boundary and shell counting)

```
INNER WALL (hard, established for all elements):
  The nuclear Zone 1/2 boundary at r ~ lambda_p = r_p/4 = 0.21 fm.
  Zone 2 cells are jammed (Maxwell-critical) and impenetrable to the
  electron's vertex-mode winding. This is a genuine physical boundary,
  not a potential — the jammed cells physically cannot be entered.
  [doc_nucleus Section 6]

SHELL COUNTING (from I_h irrep structure, not walls):
  The number of electron states per shell is determined by the I_h
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```
  irreducible representation structure of the Coulomb pressure well.
  This gives 2n² exactly for all elements, Z-independent.
  [atomic_shells.py AS1-AS7, 7/7 PASS]

NOTE ON OUTER WALL: there is no established outer wall mechanism.
  For a single proton (Z=1): the orbit radius is set entirely by the
  light-linger wave closure condition in the open Coulomb well — no
  outer wall is needed or present.
  For multi-electron atoms (Z>1): inner-shell electrons screen the
  nuclear charge, modifying the effective Coulomb gradient that the
  outer electron sees. This is a continuous potential modification, not
  a discrete wall. The outer electron's orbit is still set by the wave
  closure condition in the (now-screened) effective potential.
  The "outer wall" picture is not supported by existing scripts and is
  not required to derive 2n² shell maxima.
```

4. ROLL DIRECTION = CHARGE SIGN

4.1 The physical mechanism

The proton's (1,2) Hopf winding freezes Zone 2 cells into a specific chirality. "Positive charge" is the label for this chirality direction: the proton's inward winding. Two protons have the same chirality.

When two same-chirality sources approach, the cells between them are dragged in OPPOSING directions. The cells grind. Grinding = high pressure between them. Both sources flee the high-pressure zone. This is repulsion.

When opposite-chirality sources approach (proton + electron), the cells between them are dragged in COMPLEMENTARY directions. They MESH: the cells flow smoothly in one net direction (Bernoulli effect). Meshing = low pressure between them. Both sources follow the low-pressure zone. This is attraction.

```
SAME chirality:   GRINDING → high pressure → REPULSION
OPPOSITE chirality: MESHING → low pressure (Bernoulli) → ATTRACTION
[doc_orbit_pressure Section 1.2; doc_nucleus Section 6]
```

4.2 The electron is the proton's chirality complement

The electron has E+ winding with opposite roll to the proton's (1,2) Hopf chirality. This opposite roll is what we call "negative charge". It is not a label assigned independently -- it follows directly from the cell-meshing mechanism. The electron's roll direction is the unique E+ winding orientation that produces attraction to the proton's Zone 2/3 cell chirality.

The electron does not choose to have negative charge. The 2I double group has two E-type spinors: E+ and E-. The one that meshes with the proton is E+ (the electron). E- is the positron. Both exist as equally valid 2I irreps. The physical distinction is which one meshes with the proton's winding and which one grinds. The proton selects the labelling by fixing the chirality reference.

4.3 Measured consequence: Coulomb energy at grinding contact

```
r_grind = 2*lambda_p = 0.421 fm  [Zone 2 boundaries touch; same chirality]
Coulomb energy at r_grind: alpha_em * hbar*c / r_grind = 3.42 MeV
Observed nuclear hard core: 0.4-0.6 fm. Prediction: 0.421 fm. [MATCH]
[doc_nucleus Section 6]
```

For the proton-electron (meshing) pair: no hard core exists. Attraction is smooth all the way to Zone 1 (where the proton's impenetrable structure finally prevents further approach). This asymmetry is why stable atoms exist.

5. POSITRON COMPARISON

5.1 Identity

The positron is E- (the complex conjugate irrep of E+ in 2I; physically, roll direction reversed relative to proton Zone 2/3 chirality). It is the 2I spinor conjugate of the electron. Same mass (CPT symmetry, automatic in the torsionverse from bilateral winding symmetry), same vertex geometry as E+, same C5 character (+phi), same Born balance. E+ and E- have identical chi(C5) = phi because phi is real (self-conjugate); the Born coupling is through chi(C5) only, so n_exact and alpha are the same for both irreps. [ED19 PASS]
The sole physical difference: roll direction is reversed relative to the proton's Zone 2/3 chirality.

5.2 Positron near a proton

```
Roll: SAME as proton chirality (positron roll = proton roll direction).
Effect: GRINDING → high pressure between them → REPULSION.
```

The positron is repelled from the proton by the same cell-grinding mechanism that repels two protons. This is "like charges repel" -- now derived from geometric chirality, not assumed. The positron does not orbit the proton.

Coulomb repulsion energy at r = 2*lambda_p = 0.421 fm (the Zone 1/2 boundary radius, evaluated as the Coulomb potential at that separation):

```
alpha_em * hbar*c / (2*lambda_p) = 3.42 MeV
Note: this is not a Zone 2 contact calculation for the positron (the positron
```

has no Zone 2). It is the Coulomb potential energy evaluated at the distance where the proton's own Zone 2 boundary sits. At shorter range the repulsion continues to grow (~1/r) but without a hard Zone 2 wall, making the positron-proton repulsion softer than proton-proton at very short range.

5.3 Positron near an electron (positronium)

```
Roll: OPPOSITE to electron → MESHING → ATTRACTION.
```

The positron-electron pair attracts by the same meshing mechanism as the proton-electron pair. Positronium (e+e- bound state) forms with Bohr radius:

```
a_ps = 2 * a_0 = 1.058e-10 m
```

(twice a_0 because the reduced mass is m_e/2 vs the proton-electron case where $m_{reduced} \approx m_e$). This follows from the same pressure-gradient orbit formula with the reduced mass substitution.

Neither the electron nor the positron has a Zone 1 or Zone 2. There is no inner nuclear wall. When the e+e- pair falls to minimum separation ($r \to 0$ in the classical picture), their forward and backward E+ windings cancel directly: forward winding + backward winding = zero net winding $\to A_g$ (totally symmetric). This clean cancellation — with no Zone 1 wall to limit the approach — raises the question of what carries the released energy. Sections 5.3.1 and 5.3.2 address this.

5.3.1 ANNIHILATION AS CORPUSCLE EVIDENCE

```
Positronium annihilation constrains the nature of the electron.
In a purely classical wave model -- where E+ and E- are continuous
amplitude patterns in the medium -- destructive interference is lossless:
the medium returns to rest and no energy is released. The observed
2 × 511 keV gamma emission (PDG: m_e*c^2 per photon) requires that the
electron carries discrete energy that must be released when the configuration
cancels. This rules out the classical-amplitude model. Both QFT (discrete
quantum field excitation, by the quantization axiom) and the torsionverse
corpuscle (discrete topological winding, by Hopf winding number quantization)
satisfy the requirement. The torsionverse adds the mechanism: discreteness
follows from winding number quantization, not from a postulate. When the E+
and E- windings cancel to A_g, the topological path energy (m_e*c^2 per
corpuscle) is released as T_1g transverse waves (photon corpuscles).
The A_g symmetry of the cancellation state forces no preferred emission
axis (momentum conservation from the A_g coupling),
consistent with observation.
```

5.3.2 ANNIHILATION ENERGY BOOKKEEPING [CE5-CE6 PASS]

```
In the CM frame the E+ and E- momenta are equal and opposite in all
components. Their vector sum is zero. The total rest-mass energy of both
corpuscles (2 * m_e*c^2 = 1022 keV) is released as two 511 keV photon
corpuscles. Energy conservation requires each photon corpuscle to carry
m_e*c^2 = 511 keV [PDG2022]; the observation of two 511 keV photon
corpuscles confirms that the full winding energy is released, not retained. The direction of emission (back-to-bac
para-Ps L=0 case) follows from A_g symmetry above; the post-collision
status in non-ideal cases (orbital angular momentum, off-axis approach)
is an open item -- see open_items.txt: CORPUSCLE-ENERGY-DECOMPOSITION.
[corpuscle_energy_decomp.py CE5-CE6 PASS; see also CE1-CE2 for winding geometry]
```

Positronium lifetime:

```
tau = 2*hbar / (m_e*c^2 * alpha^5) = 124.5 ps  (para-Ps, singlet)
PDG: 125.1 ps.  Error: 0.5%  [ED14 PASS, leading-order formula]
Physical: rate at which opposite-roll E+ windings overlap and cancel.
Full geometric derivation from E+ winding cancellation overlap integral
not yet done; the leading-order formula match is closed.
```

6. THE Z > 137 COULOMB LIMIT

6.1 Mechanism

The 1s Bohr radius is $a_{1s} = a_0 / Z$. As Z increases:

```
Z = 1:   a_0      = 52,918 fm = 5.292 Angstrom  (hydrogen)
Z = 100: a_0/100  =    529 fm  (still >> r_p = 0.841 fm; outside nucleus)
Z = 137: a_0/137  =    386 fm  = lambda_e (electron Compton wavelength)
```

At Z = 1/alpha = 137.036, the 1s orbital radius exactly equals the electron's own Compton wavelength (lambda_e = hbar_c/m_e = 386 fm). This is an EXACT algebraic identity: a_0*alpha = hbar_c/m_e = lambda_e. [ED7-ED8, PASS]

In the corpuscle model the electron's winding IS its path through space: a sequence of 12 straight-line trajectories between successive icosahedral vertex impacts, with no curvature between impacts. One complete (1,2) Hopf winding = one traversal of all 12 vertices in the correct topological order. The circuit closes when the 12th impact returns the corpuscle to its starting vertex orientation.

As Z increases, the 1s orbital radius a_1s = a_0/Z shrinks. At Z = 137 it reaches exactly lambda_e (algebraic identity, ED7-ED8). Above Z = 137, no valid 12-impact Hopf circuit can close within the available circumference -- the straight-line paths between successive vertex contacts would have to shorten below the electron's own Compton-scale interaction length. The circuit fails geometrically; the orbit has no solution.

```
TORSIONVERSE MECHANISM: The orbital path (12-vertex winding circuit)
fails geometrically -- the winding can no longer close. The corpuscle has
no valid path and impacts Zone 2 from outside. Zone 2 cells are locked in
the proton's chirality; the impact is a kinetic energy transfer through
the jammed cells to the Zone 1 boundary. There, the u-quark winding
absorbs the energy and executes the antipodal vertex bounce:
  u-quark (T_1u at vertex v) -> d-quark (T_2u at antipodal vertex -v)
because chi(T_1u, C5^4) = -1/phi = chi(T_2u, C5) -- exact, geometric.
Proton (uud) becomes neutron (udd). The corpuscle exits Zone 2 as a freed
winding. Zone 2's proton chirality acts as a chirality mirror: the
incoming E+ winding exits as E- (opposite roll). A positron is an E-
with Compton-scale Zone 2 interaction range (it grinds with Zone 2); this
freed E- has no orbital structure at that scale and therefore interacts
only weakly -- it propagates without orbital structure, which is the
defining property of a neutrino mode. The formal irrep assignment of the
electron neutrino is in [doc_leptons]; the key point here is that the
same E- winding that is a positron at Compton-scale orbital energies
becomes neutrino-like when stripped of that structure.
The CG algebra requires exactly this:
  T_2g x E+ -> I52 -> T_1g + E-  [WI1 reversed, exact]
The electron never enters Zone 1 or Zone 2. The whole event is kinetic.
[weak_quark_flip.py: antipodal vertex bounce; weak_interaction_cg.py WI1]
Z_crit = 1/alpha emerges from the corpuscle circuit geometry -- derived
from the same icosahedral vertex structure that produces alpha, not
assumed from the Dirac equation.

STANDARD QED (same threshold, implicit mechanism): The Dirac equation
predicts the same Z_crit = 1/alpha -- the 1s bound state energy dives
below -m_e*c^2 at this point (atomic collapse / supercritical Coulomb
vacuum; [Pomeranchuk1945]; [Gershtein1969]; confirmed in graphene
[Shytov2007]). The end-state matches ordinary nuclear electron capture; the
entry condition differs (geometric circuit failure vs energy threshold).
The torsionverse supplies the mechanism QED leaves implicit.
```

NOTE: the Jobson cell scale L_J = 9.93e-3 fm is not the relevant UV cutoff here. L_J is 40,000x smaller than lambda_e. The atomic system's UV limit is set by the electron's Compton wavelength, not the cell size. Z_crit = 1/alpha = 137.036 is the cell-geometry limit because alpha is the ratio of the electron's Born coupling to the orbital wave condition:

```
a_0 = lambda_e / alpha  =>  Z_crit = a_0 / lambda_e = 1/alpha.
```

[doc_nucleus Section 6.3; electron_doc.py ED7-ED9]

6.2 Alpha as the cell-to-orbital scale ratio

This limit exposes the deepest connection: the fine structure constant alpha is not only the electron's Born coupling strength -- it encodes the ratio of Compton scale to orbital scale (lambda_e / a_0), exact from the identity a_0 * alpha = lambda_e derived in Section 6.1. The atomic system lives at scales L_J << r << a_0, i.e., in the window 1 << N_J_e = 38870 << N_J_orbital (N_J_orbital = a_0/L_J ≈ 5×10■). At Z = 137, the orbital collapses to the Compton scale (lambda_e), erasing this window. Alpha's numerical value sets the window in which stable atoms can exist.

7. SUMMARY: THE FOUR FORCES AND THE ROLL QUESTION

```
FORCE COMPARISON (hydrogen Z=1, ground state E■ = -13.6 eV as reference):

Force        Torsionverse source            Role                      %
-----------  ------------------------------ ------------------------  ---------
Pressure     Zone 3 pressure gradient       Creates the potential     ~100%
             (Coulomb well from proton)     well -- without it,       binding
                                            no orbit is possible      energy

Light linger Wave asymmetry in the same     Selects WHICH radius      ~100%
             pressure gradient              is stable: the Bohr       orbital
             (wave closure condition)       radius is a linger-set    position
                                            resonance, not a
                                            classical force balance.
                                            Lamb shift (QED) = 0.00003%
                                            correction beyond this
                                            first-order closure.

Roll         Zone 2→Zone 3 cog coupling      Sets orbital SHAPE        ~0.0003%
             via gluon-antinode contact     (l, m_l): fine structure  energy
                                            splitting per shell         (alpha^4/E_1)

Wall         Zone 1/2 hard boundary +       Inner: prevents r→0        0% energy
             I_h irrep counting (AS1-AS7)   Counting: 2n² per shell   counting

NOTE: Pressure and light linger are inseparable. Pressure creates the
gradient; linger is how a wave particle reads that gradient to select
a stable radius. The Bohr radius = the radius where the electron's
winding circuit closes exactly once in one orbital period (the wave
resonance condition in the pressure-modified medium). Classical force
balance alone does not determine this -- the wave nature does.
Both are 100% contributors to different aspects of the same orbital.

ROLL = CHARGE SIGN:
  Electron (E+, meshing with proton) = negative charge: DERIVED [Section 4]
  Positron (E-, grinding with proton) = positive charge: DERIVED [Section 5]

POSITRON vs ELECTRON DIFFERENCES:
  Mass:    Identical (CPT, from bilateral winding symmetry) [CLOSED]
  Charge:  Opposite (roll direction relative to proton) [CLOSED, Section 4-5]
  Orbit:   Positron does NOT orbit proton (grinds); electron does (meshes)
  Annihil: Positron + electron → A_g (photons, clean, no Zone 1 wall) [CLOSED]
           Antiproton + proton → Zone 1 gluons released as pions (messy) [doc_nucleus]

THE ROLL ANSWER:
  The electron must roll the way it does relative to the proton because its
  E+ winding (forward roll) is the specific 2I spinor whose chirality
  complements the proton's (1,2) Hopf Zone 2 winding. If the electron had
  the opposite roll (= positron), it would not form a stable orbit around
  the proton at all. The universe selects which E-type spinor is "electron"
  and which is "positron" by the proton's chirality fixing the reference.
  CP violation (slight asymmetry in matter vs antimatter abundance) may
```

trace to a slight imbalance in the initial population of E+ vs E- roll states in the cosmological medium. (speculative, not yet derived)

8. OPEN ITEMS AND RECENT CLOSURES

INNER-SHELL SCREENING: [PARTIAL — IS1-IS9 PASS, orbital penetration open]
The electron creates a Coulomb pressure field V(r) = alpha*hbar_c/r from displacing cells in its wake (fractional displacement alpha per contact, Section 3.1). Gauss's law applied to this 1/r^2 field gives the shell theorem exactly:

```
  sigma_inner  = 1.00  (inner electron fully screens; IS1-IS2 PASS)
  sigma_outer  = 0.00  (outer electron gives no screening; IS1-IS2 PASS)
  sigma_same_1s= 5/16  (same-shell 1s: from 1s radial density, not uniform
                        sphere — corpuscle path IS the density; IS4 PASS)
  sigma_classic= 1/2   (uniform sphere upper bound; IS3 PASS)
```

The 5/16 result bounds Slater's empirical 0.30-0.35 correctly (IS6 PASS). He ground state energy with sigma=5/16: -77.49 eV vs measured -79.01 eV, 1.92% error (IS5 PASS; expected — leading-order variational estimate).
Script: analysis/atomic/inner_shell_screening.py [9/9 PASS]

STILL OPEN: orbital penetration — the sigma=0.85 Slater coefficient for n-1 shell electrons. This accounts for the fraction of time the 2s corpuscle spends inside the 1s boundary (the 2s wavefunction has a node at 2a_0/Z but nonzero density below a_0/Z). Fraction ~ 15%, giving sigma ~ 0.85. Deriving this from torsionverse requires the 2s-vs-1s winding-path overlap integral. See open_items.txt: ELECTRON-INNER-SHELL.

CP VIOLATION: (speculative; mechanism identified but not yet derived)
The torsionverse medium is built from light: the (1,2) Hopf winding is a light-mode lock. When Zone 2 first froze in our local region, it locked into one of two equally valid chiralities. Which chirality it chose was decided by whatever local fluctuation triggered the first freeze — not by a fundamental universal asymmetry.

Consequence: the matter-antimatter imbalance is our LOCAL condensation bias, not a property of the universe as a whole. Causally-disconnected regions that condensed with the opposite chirality would be entirely antimatter-dominated. We cannot easily detect them: light from antimatter stars is indistinguishable from matter-star light. The boundary between regions would show a pion annihilation zone — a gamma-ray excess along certain directions.

The observed η ≈ 6×10^-10 (baryon-to-photon ratio) may therefore not be a number to derive from first principles — it reflects the accident of our local freeze. The torsionverse "predicts" only that η is nonzero and region-dependent; its exact value is initial-condition noise.

CHIRALITY CASCADE:
The local proton chirality is not chosen at the atomic scale -- it inherits from a cascade of medium currents: cosmic web filament → galactic center (depleted zone, pressure-gradient driver of galactic rotation) → solar nebula spin → Zone 2 freeze. The galactic center acts as a chirality AMPLIFIER: once a rotation sense is established at galaxy scale it propagates coherently downward via the near-incompressible medium (K/G = 30.25; bulk-to-shear modulus ratio of the torsion medium, derived in [doc_jobson_cell]) to the Zone 2 freeze of individual protons. The chain terminates at the cosmic web scale, where the original filament angular momentum was set by quantum fluctuations in the initial medium condensation -- random in sign, region by region. No universal primary driver exists; the chirality of our matter is inherited from our cosmic web filament's accident of formation.

Closed items (ED15-ED19):
- Spin-orbit coupling: ΔE(2p) = alpha^4*m_e*c^2/32 = 45.3 μeV, measured 45.35 μeV, 0.15% [ED15 PASS]
- g-factor / Zone 3 effect on electron: the proton Zone 3 co-rotation couples to the electron spin via the effective magnetic moment (g_p = 5.5857), producing the hydrogen hyperfine splitting (21 cm line). Computed: (4/3)*alpha^4*m_e*(m_e/m_p)*g_p = 1421.0 MHz, measured 1420.4 MHz, 0.04% [ED16 PASS]. Zone 3 co-rotation does NOT modify the free-space g-factor (2.00232); it creates the spin-energy splitting, which is already captured.
- Zone 3 co-rotation radial falloff: derived as 1/r^3 (Lense-Thirring). E_Z3(r_p) = alpha*hbar_c/r_p = 1.71 MeV (onset); E(r)/E(2r) = 8 (exact).

```
  [ED17-ED18 PASS; entanglement_doc.py EP1-EP2]
- Positron Born balance: chi(E-,C5) = chi(E+,C5) = phi (phi is real,
  self-conjugate); Born coupling through chi(C5) gives same n_exact (the Born-
  balance integer from [doc_alpha] J17/J24) -> same
  alpha for E- as for E+. [ED19 PASS]
```

COMPANION SCRIPTS (reproducibility):

```
analysis/demos/electron_doc.py    -- 19/19 PASS  [STANDALONE -- no external deps]
analysis/demos/orbit_doc.py       -- [OD6, OD11-OD14; see doc_orbit_pressure]
analysis/gravity/pressure_isobar_orbit.py -- [PO2; Section 3.1, Bohr radius]
analysis/quantum/cell_rotation_propagation.py  -- [RP1-RP4; Section 3.3]
analysis/demos/entanglement_doc.py -- [EP1-EP2; Sections 3.3, 8]
analysis/nuclear/atomic_shells.py  -- [AS1-AS7; Section 3.4]
analysis/quantum/weak_quark_flip.py -- [Section 6.1]
analysis/quantum/weak_interaction_cg.py -- [WI1; Section 6.1]
analysis/atomic/inner_shell_screening.py -- [IS1-IS9; Section 8, inner-shell screening]
analysis/atomic/corpuscle_energy_decomp.py -- [CE1-CE2; Section 2.3], [CE5-CE6; Section 5.3]
Repository: https://doi.org/10.5281/zenodo.22105622
Code:        https://github.com/Destraag/torsionverse_public
```

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(See also entanglement_doc.py EP1-EP2, cited in Section 3.3)

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